A Hybrid ITTA–TLBO Algorithm for Optimization

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Abstract

In this work, we present a hybrid optimization algorithm based on a **Low-Level Teamwork Hybrid** cooperation between the **Improved Tiki-Taka Algorithm (ITTA)** and **Teaching-Learning Based Optimization (TLBO)**. Unlike sequential hybridization, this approach integrates both mechanisms within each iteration, combining the adaptive exploratory capacity of ITTA with the strong local exploitation ability of TLBO. We detail the algorithmic steps, the main mathematical equations, and highlight the balance achieved between exploration and exploitation.

1 Introduction

Optimization is a key field in operations research and artificial intelligence. Hybrid metaheuristics aim to exploit the complementary strengths of different algorithms. In this work, we propose a **Low-Level Teamwork Hybrid** scheme where ITTA and TLBO cooperate within each iteration. This combination improves convergence quality and robustness against local minima.

2 Methodology: Proposed Algorithm

The algorithm follows these steps:

2.1 Initialization

The positions of players and balls are generated randomly as:

$$P = \begin{bmatrix} p_{11} & \cdots & p_{1d} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nd} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & \cdots & b_{1d} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nd} \end{bmatrix} \tag{1}$$

2.2 Evaluation

Compute the fitness values and select the best players (top 10%, minimum of three).

2.3 Iteration Loop

At each iteration, the following steps are executed:

- Update of the balls:

$$b_i' = \begin{cases} rand(b_i - b_{i+1}) + b_i, & rp > prob_{lose} \\ b_i - (c_1 + rand)(b_i - b_{i+1}), & rp \leq prob_{lose} \end{cases} \tag{2}$$

- Update of the factors:

$$r = c_2 \frac{iter}{max_iter^2}, \quad s = c_3 \sqrt{1 - \frac{iter}{max_iter}} \tag{3}$$

- Update of the players' positions:

$$p_i' = p_i + rand \cdot r(b_i' - p_i) + rand \cdot s(h - p_i) \tag{4}$$

- Teacher phase (TLBO):

$$X_i = X_i + r_i(T - T_f M), \quad M = \frac{1}{N} \sum_{i=1}^{N} X_i \tag{5}$$

- Learner phase (TLBO):

$$X_i = \begin{cases} X_i + r_i(X_k - X_i), & F(X_k) < F(X_i) \\ X_i + r_i(X_i - X_k), & otherwise \end{cases} \tag{6}$$

2.4 Stopping Criterion

These steps are repeated until the maximum number of iterations is reached.

2.5 Final Result

The algorithm returns the best solution (optimal trajectory) and its associated fitness value.

3 Pseudo-code

Algorithm 1 Hybrid ITTA–TLBO Algorithm

```
Initialize positions P and B
Evaluate initial fitness and select the best players
for iter = 1 to max_iter do
    Update B according to (2)
    Compute r and s according to (3)
    Update P according to (4)
    Apply TLBO Teacher Phase according to (5)
    Apply TLBO Learner Phase according to (6)
    Re-evaluate fitness and update the best players
end for
Return the best solution found
```

4 Discussion

This **low-level cooperative hybridization** combines:

- the adaptive exploration of ITTA;
- the local exploitation of TLBO;
- an effective balance between exploration and exploitation.

As a result, the algorithm achieves better convergence performance and increased robustness.

5 Conclusion

The hybrid ITTA–TLBO algorithm is a promising approach for solving complex optimization problems. Future perspectives include applying it to real-world domains such as cloud scheduling, energy optimization, or Path Planning.